

Yuliang Peng

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EDUCATION

University of Maryland, College Park

M.S. in Applied Machine Learning

GPA: 3.8/4.0, May 2027

B.S. in Computer Science

GPA: 3.6/4.0, May 2024

Relevant Coursework: LLM Systems, LLM Methods & Applications, Cloud Computing, Computer Vision, Deep Learning

WORK EXPERIENCE

ICBC USA — *Software Engineer Intern* | Los Angeles, CA | Jun 2026 - Aug 2026

- Built a payment reconciliation validation workflow that cross-checks posting records, settlement files, and exception reports to detect missing transactions, duplicate entries, amount mismatches, and failed payment states.
- Automated exception reporting and root-cause tracking for reconciliation breaks, improving payment-operation traceability and accelerating downstream investigation for reporting teams.
- Translated manual review steps, compliance constraints, and anomaly-detection use cases into technical requirements for AI-assisted internal workflow tooling.

ModeSens — *Software Engineer* | Remote, USA | Apr 2025 - Sep 2025

- Built backend ingestion services for product feeds across 30+ global retailers, standardizing product metadata, images, pricing, and retailer-specific schemas into Python and AWS-based data workflows.
- Engineered production data pipelines with CI/CD automation, validation checks, and failure-handling logic, reducing feed-processing failures by 20% and improving data reliability for analytics and recommendations.
- Developed an image-matching system using perceptual hashing, SSIM, and Selenium to detect duplicate and variant products, reducing manual review effort by 80% and improving recommendation accuracy by 25%.

Bank of China — *Software Engineer* | Los Angeles, CA | Jul 2024 - Apr 2025

- Built backend services for transaction monitoring systems, supporting risk review, compliance workflows, exception handling, and operational reporting across financial records.
- Resolved 70+ production incidents across services, reports, and data pipelines through log analysis, root-cause isolation, and pipeline fixes, reducing blocking incidents by 20%.

PROJECTS

[InferRoute](#) | LLM Inference Gateway & Control Plane | FastAPI, Redis, PostgreSQL, Docker

- Built a policy-driven LLM gateway that routes requests across OpenAI, Gemini, and local model backends using task type, context length, provider health, latency history, token-cost estimates, and SLO-aware routing rules.
- Engineered inference reliability with Redis caching, request deduplication, timeout budgets, circuit breakers, retry/backoff logic, provider failover, and degraded-provider recovery tests.
- Built PostgreSQL-based tracing and benchmark dashboards for p50/p95/p99 latency, token usage, cost per request, cache-hit rate, routing-policy comparison, and failover recovery analysis.

[Quant.ai](#) | Quant Research & Backtesting Agent Platform | FastAPI, React, TypeScript, Pandas, NumPy

- Built a full-stack quant research platform with a FastAPI backtesting engine and React dashboard for natural-language strategy definition, walk-forward validation, experiment comparison, and risk analysis.
- Designed an LLM tool-calling workflow and vectorized simulation engine for market data ingestion, indicator computation, regime-aware backtesting, ATR position sizing, drawdown controls, and structured risk reports.

[TabIt](#) | Intelligent Chrome Workspace Extension | React, TypeScript, Chrome Extension MV3

- Built a Chrome Extension to organize browser tabs into semantic groups with real-time clustering, archived sessions, and search.
- Implemented Web Worker processing and incremental client-side state synchronization, achieving sub-100ms latency for 100+ tabs.

PUBLICATION

International Cargo Charter Operations and Preighter Performance | *Transport Policy*, Vol. 184, Article 104173, 2026

- Co-authored Python route-network analysis of CAAC cargo approvals. DOI: 10.1016/j.tranpol.2026.104173.

SKILLS

ML / Data: PyTorch, TensorFlow, Hugging Face, NumPy, Pandas, Selenium

Backend / Infrastructure: FastAPI, REST APIs, Docker, Redis, PostgreSQL, AWS, CI/CD, Linux, Git

Languages: Python, C++, Java, JavaScript/TypeScript, SQL | Spoken: English, Chinese, Japanese